<u>CS1005 - DISCRETE STRUCTURES</u>

Quiz No. 2
Name:
Roll No:
Section: D
Total Marks: 10

Q1. For each of the following formulas write a formula *f* (using quantifiers) expressing the formula, find a formula in negation normal form equivalent to ~*f*, and express the meaning of the negation in words. **(3 Marks)**

(a) Every number x has a square root. (Do not use the square root symbol, use only multiplication)

(b) For some x and y, $x < y$, and $x^3 - x > y^3 – y$

(c) For every x and y, if $x^3 + x - 2 = y^3 + y - 2$, then $x = y$

NOTE: Negation Normal Form means negating the formula and pushing the ~ symbol inward until it is only in front of the variables/predicates.

Q2. Let *S* be the set of students at your school, let *M* be the set of movies that have ever been released, and let V(s, m) be "student s has seen movie m." Rewrite each of the following statements in English: **(2 Marks)**

(a) $\forall s \in S, \exists m \in M$ such that $V(s, m)$

(b) $\exists s \in S\ \exists t \in S$ such that $s \neq t$ and $\forall m \in M,\ V(s,m) \rightarrow V(t,m)$

Q3. Let the domain be all people in a university.

$S(x)$: "x is a student"

$H(x, y)$: "x has worked on a project with y"

$C(x)$: "x studies Computer Science"

Translate the statements a, b and c into logical form using quantifiers, connectives and the given predicates: **(5 Marks)**

(a) Every Computer Science student has worked on a project with at least one student.

(b) There is a student who has worked on a project with every student.

(c) There exists a student who has never worked on a project with any Computer Science student.

(d) $\forall x(C(x) \rightarrow S(x))$, if this is true then can we conclude that every student is a Computer Science student?